

Use-Case Flow – UC-01 Record Monthly Consumption & Calculate CO_{2e}

Problem #61 | Household Carbon Footprint & Target Tracker

Preconditions

1. Resident is authenticated and has access to the household record.
2. A valid month is selected for recording.
3. Required CO_{2e} conversion factors are available.

Postconditions

1. Electricity, gas, and commute values are stored for the selected month.
2. Monthly CO_{2e} is calculated and stored/displayed.
3. Historical emissions and target-progress information are updated.

Main Success Scenario

1. Resident selects “Record Monthly Consumption.”
2. System displays the monthly consumption form.
3. Resident enters electricity (kWh), gas consumption, and commute kilometres for the selected month.
4. System validates that all required values are present and numeric.
5. System obtains/validates the required conversion factors.
6. System calculates total monthly CO_{2e}.
7. System stores the monthly record and calculated CO_{2e}.
8. System displays the CO_{2e} result and updated history/target progress.
9. Use case ends successfully.

Alternate Flow – Invalid Monthly Data

- 4a. System detects a missing, non-numeric, or invalid value.
 - 4a1. System displays a validation message identifying the invalid field.
 - 4a2. Resident corrects the value and resubmits the form.
 - 4a3. If valid, the flow resumes at Step 5; otherwise, the record is not saved.